

UTKARSH SHEEL ANAND

 utkarshsheel.netlify.app |  utkarshsheel@gmail.com | Ottawa, ON

EDUCATION

Carleton University

B.Eng. Aerospace Engineering (Space Systems Design)

Ottawa, ON

GPA: 10.43/12 (A-) | Dean's Honour List

PROFESSIONAL EXPERIENCE

National Research Council Canada

Co-op Student, Aerospace Research

Ottawa, ON

May 2023 – Aug 2024

- Optimized the aerodynamic shape of S-duct intakes for boundary-layer-ingesting engines using CFD, improving flow uniformity for next-generation propulsion integration.
- Designed and verified a low-cost anechoic chamber compliant with ISO 3745 and ISO 26101, supporting acoustic test-setup validation.
- Engineered pin-fin and micro-structured heat sink geometries for hybrid-electric aircraft battery cooling, improving heat transfer coefficient by up to 85% and reducing mass by 27% versus a baseline plate-fin design.
- Co-authored “*Experimental Investigation of Enhanced Heat Sinks for Hybrid Electric Aircraft*” (ASME Turbo Expo 2025, GT2025-151174), leading test-rig setup, instrumentation, data acquisition, and thermal analysis.

Multi-Domain Laboratory, Carleton University

Research Assistant

Ottawa, ON

May 2022 – May 2023

- Developed a MATLAB image-processing pipeline to detect and track whale blow behavior for environmental sensing research.
- Integrated software and hardware components for field-deployable environmental sensing prototypes.

TECHNICAL PROJECTS

Error-State Kalman Filter for Sensor Fusion | *Python, NumPy*

- Implemented a 15-state Error-State Kalman Filter from scratch, fusing 200 Hz IMU propagation with asynchronous GPS and odometry corrections for 6-DOF pose estimation.
- Achieved 3.9 cm 3D position RMSE over 36,000+ samples on the EuRoC MAV benchmark using quaternion error-state correction and Joseph-form covariance updates.

Action-Chunking Transformer Policy for Robotic Manipulation | *PyTorch, Hugging Face LeRobot*

- Trained a transformer-based behavior-cloning policy on 45 teleoperated demonstrations to control a physical SO-101 arm on a pick-and-place task.
- Achieved zero failures on in-distribution placements (8 clean, 2 near-miss) and 57% clean success across all 14 real-robot rollouts, surfacing generalization limits outside the demonstrated workspace.

PPO Reinforcement Learning for Continuous Control | *PyTorch, Stable-Baselines3, MuJoCo*

- Trained and evaluated PPO agents across 8 runs (27M+ environment steps) on Hopper, Walker2d, Ant, and Humanoid locomotion tasks using a standardized checkpoint-selection protocol.
- Ran controlled ablations on learning rate, seed, and rollout length, reaching top evaluation scores of 6,008 on Walker2d and 4,451 on Humanoid.

ZenohFleet: Cross-Platform Robotics Middleware & Vision Pipeline | *ROS 2, Zenoh, YOLOv8*

- Built a two-machine messaging pipeline bridging a ROS 2 (Ubuntu) robot to a ROS-free macOS ground station via zenoh-bridge-ros2dds.
- Decoded ROS 2 CDR messages natively on macOS, ran real-time YOLOv8 object detection, and returned serialized velocity commands over a closed-loop LAN pipeline.

Autonomous Quadrotor Navigation | *PX4, ROS, Gazebo*

- Developed a PX4/ROS/Gazebo guidance stack combining PD trajectory tracking with LiDAR-based Artificial Potential Field obstacle avoidance.
- Demonstrated stable autonomous takeoff, hover, and circular-path tracking with real-time reactive avoidance of moving obstacles.

ROS 2 Differential-Drive Mobile Robot | *ROS 2, Gazebo, RViz*

- Built a mobile robot from a CAD-accurate URDF to a physics-based Gazebo simulation with a validated TF tree and `ros2_control` diff-drive controller.
- Delivered a reproducible bring-up pipeline with real-time teleop and fused odometry, laying groundwork for navigation and SLAM integration.

Weather Satellite Mission Design | *MATLAB, Satellite Toolbox*

- Designed a Pre-Phase-A Sun-synchronous LEO weather satellite mission, sizing power, thermal, ADCS, propulsion, and communications subsystems per SMAD III methodology.
- Closed S-band QPSK link and power budgets (~ 80 min/day access, 12.3 m^2 solar array, 2.6 kWh battery) and sized bipropellant mass for station-keeping and deorbit.

Spacecraft Attitude Dynamics & Control | *MATLAB/Simulink*

- Simulated 6-DOF reaction-wheel spacecraft attitude dynamics with quaternion-based closed-loop control for Nadir pointing.
- Built custom guidance, control, and Star Tracker blocks; validated controller robustness under drag, SRP, gravity-gradient, and magnetic disturbance torques across four gain-tuning strategies.

Low-Cost Gas Chromatograph – Europa Mission Payload Concept | *Arduino, MATLAB*

- Designed and built a bench-top gas chromatograph using a packed silica-gel column, TGS-813 gas sensor, and Arduino-based data acquisition as a candidate instrument for planetary sample analysis.
- Achieved clear analyte separation of butane with a $\sim 5.4 \text{ s}$ retention time, validating a low-cost, flight-concept-ready chromatography approach.

TECHNICAL SKILLS & CERTIFICATIONS

Robotics & Autonomy: ROS / ROS 2 (Humble), PX4, MAVROS, Gazebo, Zenoh, TF, LiDAR-based obstacle avoidance

Robot Learning & Computer Vision: PyTorch, Hugging Face LeRobot (ACT), Stable-Baselines3 (PPO), Gymnasium/MuJoCo, YOLOv8, OpenCV

Guidance, Navigation & Control: Spacecraft ADCS, Error-State Kalman Filtering, PX4 EKF2, PD Control, Reaction Wheels, Magnetorquers

Satellite & Mission Systems: STK (Level 2, Level 3 RF), MATLAB Satellite Toolbox, Orbital Mechanics, Power/Link Budgeting, GNSS (RTKLIB, GLAB)

Aerospace Design & Simulation: CFD, XFLR5, ANSYS, SolidWorks, CATIA, MATLAB/Simulink, Simscape Multibody

Programming & Tools: Python, C++, MATLAB, NumPy, Arduino, Raspberry Pi, Linux, Git, LabVIEW

Certifications: STK Level 2 & Level 3 RF Communications; Coursera Machine Learning Specialization (Neural Networks, CNNs, Supervised/Unsupervised Learning)